

Variation, Untangling, and Genotyping

Njagi Mwaniki

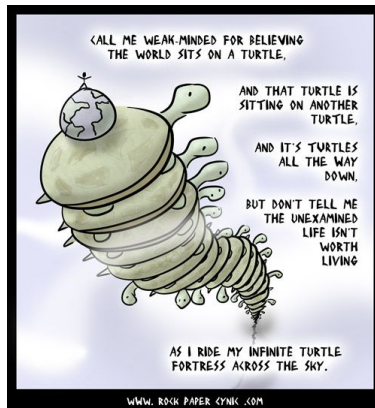
EMBO Pangenomics

13th June 2026

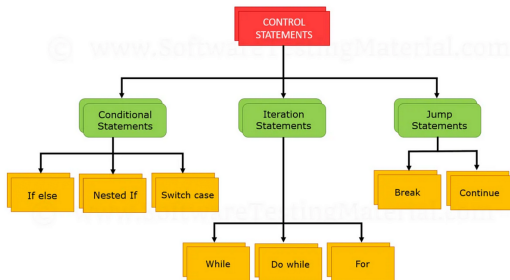


- 1 Background
- 2 Graph Decomposition
- 3 Variant Calling
 - Allele Traversal
 - MultiAllelic Sites
- 4 Untangling Repetitive Regions

Where this belongs



Control Flow Graphs



1

¹Figure from Mahdin Academy Medium

Control Flow Graphs

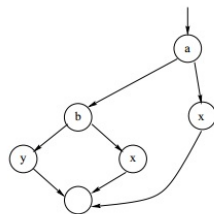
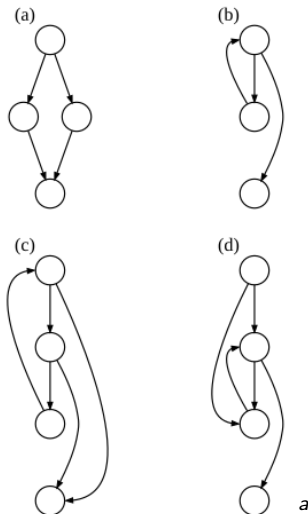


Figure 2: Compound logic.

a

^aFigure from DZone

Control Flow Graphs

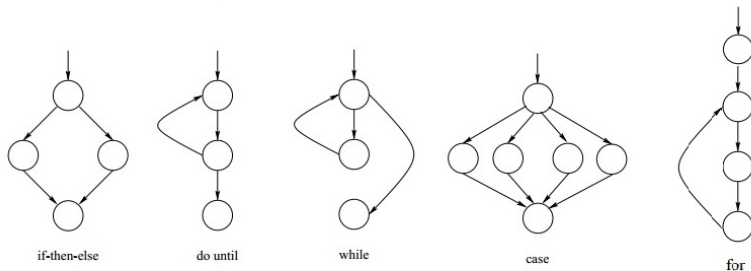


Figure 1: Flow graph representation.

1

¹Figure from dzone

Control Flow Graphs

CFGs do not map well to variation graphs

Some properties of CFGs

- distinct *start* and *end* vertices
- directed

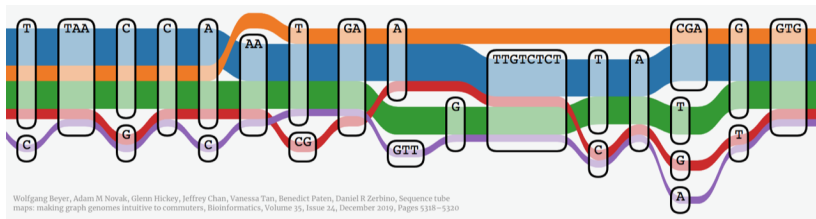
Inspired by ideas in compiler theory (Johnson et al. 1994), we developed

- an algorithm (SimpleCycleEquiv) and,
- a tool (povu).

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Pangenome Representation - Sequence Tube Map

Sequence Tube Map

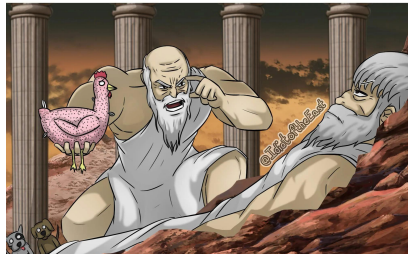


Definitions

Definitions are hard, so we shall avoid them today.

You can't just define man as
"featherless bipeds"! I mean look at
this chicken I just plucked! Does this
featherless biped look like a man to you?!

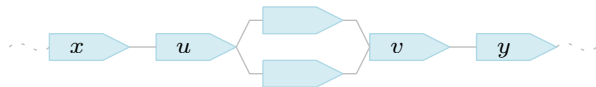
THINK PLATO THINK!



Bidirected and Biedged Graph

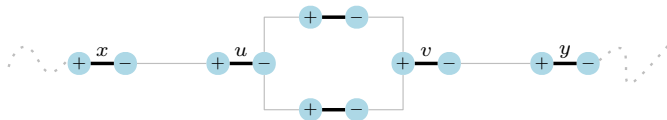
Bidirected Graph

Each edge (vertex) has two sides



Biedged Graph

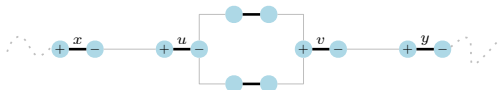
Split each vertex into two; connect them with a black edge. Previous edges are grey edges.



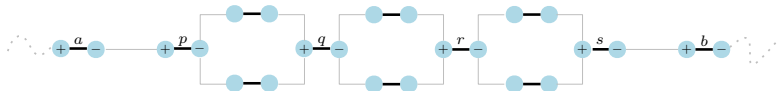
Flubble

We shall use flubbles as we would bubbles.

(a)



(b)



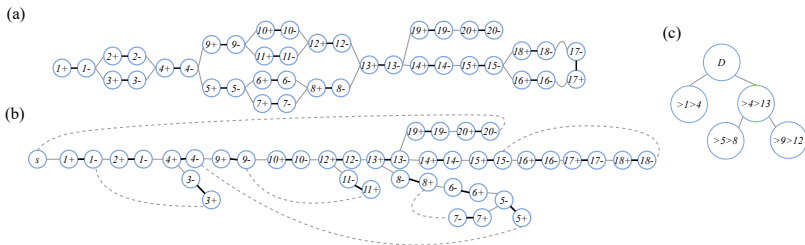
Trees

A type of graph which may not have cycles.
There can only exist one path between any two nodes.

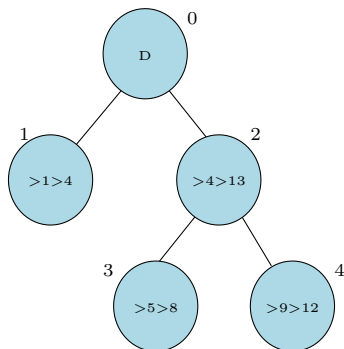
Properties:

- nodes have a parent-child relationship
- single root node
- have leaves

Trees



Pangenome Variation Structure Tree (PVST)



H	0.0.3	.	.
D	0	.	1,2
F	1	>1>4	.
F	2	>4>13	3,4
F	3	>5>8	.
F	4	>9>12	.

.flb format

Pangenome Variation Structure Tree

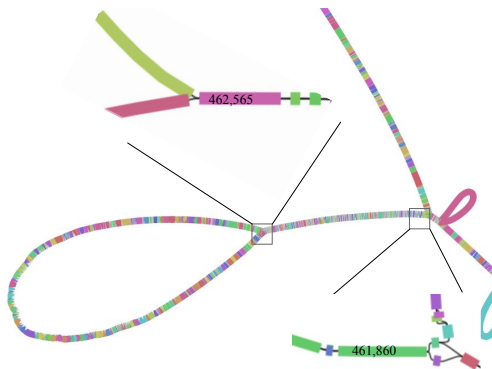
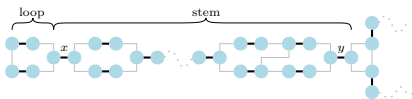
We denote using T_v

Properties:

- a rooted tree with root a dummy vertex.
- Its non-root vertices correspond exactly to the flubbles.
- Parent-child edges in T_v represent the nesting relation between flubbles
- Flubbles are non-overlapping

Hairpin Inversion

1-edge-disconnectable. A kind of structural variation, i.e. affects ≥ 50 bp.



Summary/Too Long Didn't Listen (TLDL)

- Decompose the graph into non-overlapping regions
- Call variants on the leaves
- Decomposition algorithm is conceptually simple

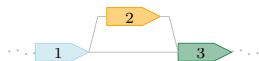
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Encoding Allele Traversal

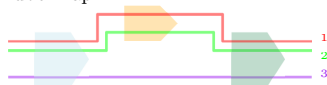
How we encode allele traversal in povu.

- 0 no traversal
- 1 traverse in forward orientation
- 2 traverse in reverse orientation

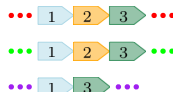
Flubble



Tube Map



Linear Projection

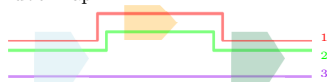


Encoding Allele Traversal

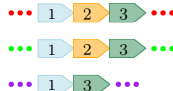
Flubble



Tube Map



Linear Projection



$$\text{Ref} = H_3$$

$\bar{S}$	1	2	3
H_1	1	1	1
H_2	1	1	1
H_3	1	0	1

- 0 no traversal
- 1 traverse in forward orientation
- 2 traverse in reverse orientation

XOR Truth Table

e**X**clusive OR ($\oplus$)

Input 1	Input 2	Output
T	T	F
T	F	T
F	T	T
F	F	F

Input 1	Input 2	Output
1	1	0
1	0	1
0	1	1
0	0	0

XOR on the Haplotypes

$\bar{S}$	1	2	3
H_1	1	1	1
H_2	1	1	1
H_3	1	0	1

H_1	H_3	H_x
1	1	0
1	0	1
1	1	0

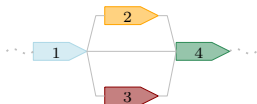
H_2	H_3	H_x
1	1	0
1	0	1
1	1	0

H_1	H_2	H_x
1	1	0
1	1	0
1	1	0

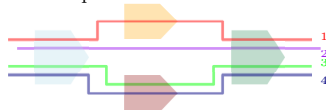
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All v All Hap Comparison

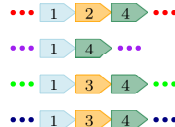
Flubble



Tube Map



Linear Projection



$\bar{S}$	1	2	3	4
H_1	1	1	0	1
H_2	1	0	0	1
H_3	1	0	1	1
H_4	1	0	1	1

All v All Hap Comparison to Genotypes

$\bar{S}$	1	2	3	4
H_1	1	1	0	1
H_2	1	0	0	1
H_3	1	0	1	1
H_4	1	0	1	1

H_1	H_3	H_x
1	1	0
1	0	1
0	0	0
1	1	0

H_2	H_3	H_x
1	1	0
0	0	1
0	1	1
1	1	0

H_1	H_2	H_x
1	1	0
1	0	1
0	0	0
1	1	0

H_3	H_4	H_x
1	1	0
0	0	0
1	1	0
1	1	0

All v All Hap Comparison to Genotypes

H_1	H_3	H_x	H_2	H_3	H_x	H_1	H_2	H_x	H_3	H_4	H_x
1	1	0	1	1	0	1	1	0	1	1	0
1	0	1	0	0	1	1	0	1	0	0	0
0	0	0	0	1	1	0	0	0	1	1	0
1	1	0	1	1	0	1	1	0	1	1	0

$\bar{S}$	1	2	3	4
H_1	1	1	0	1
H_2	1	0	0	1
H_3	1	0	1	1
H_4	1	0	1	1

Ref $\{\{H_2\}\}$

Variant sets $\{\{H_1\} \{H_3, H_4\}\}$

All v All Hap Comparison to Genotypes

H_1	H_3	H_x	H_2	H_3	H_x	H_1	H_2	H_x	H_3	H_4	H_x
1	1	0	1	1	0	1	1	0	1	1	0
1	0	1	0	0	1	1	0	1	0	0	0
0	0	0	0	1	1	0	0	0	1	1	0
1	1	0	1	1	0	1	1	0	1	1	0

$\bar{S}$	1	2	3	4
H_1	1	1	0	1
H_2	1	0	0	1
H_3	1	0	1	1
H_4	1	0	1	1

Ref $\{\{H_2\}\}$

Variant sets $\{\{H_1\} \{H_3, H_4\}\}$

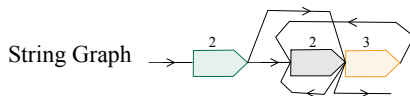
In VCF

...	Ref	Alt	...	H_1	H_2	H_3	H_4
...	S_0	S_1, S_2	...	0	1	2	2

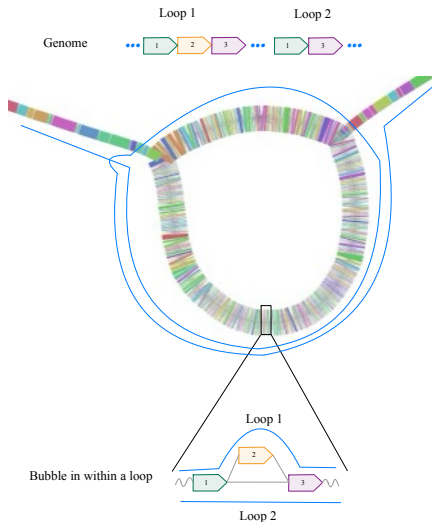
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Tangling (Single Genome)

Genome ...  ...   ...  ...  ...  ...  ...

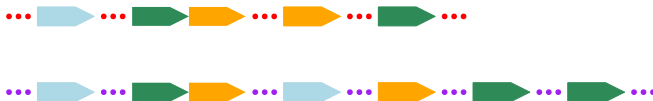


Tangling (Single Genome LPA)

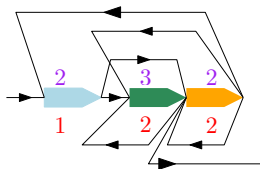


Tangling (Multiple Genomes)

Genomes



Graph

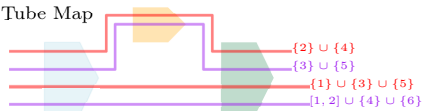


References in a RoV (flubble)

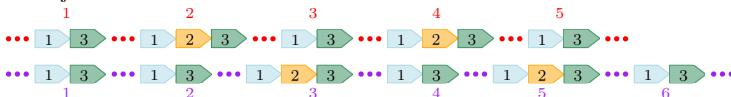
Flubble



Tube Map



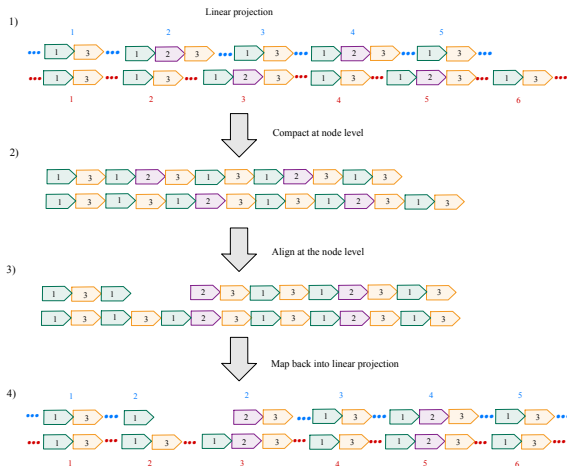
Linear Projection



- **Allele Traversal:** An *uninterrupted* sequence of steps that a reference takes from the start to the end of a single RoV
- **Itinerary:** A sequence of allele traversals for a reference in a single RoV.

subtext: in an itinerary, the sequence of steps is interrupted

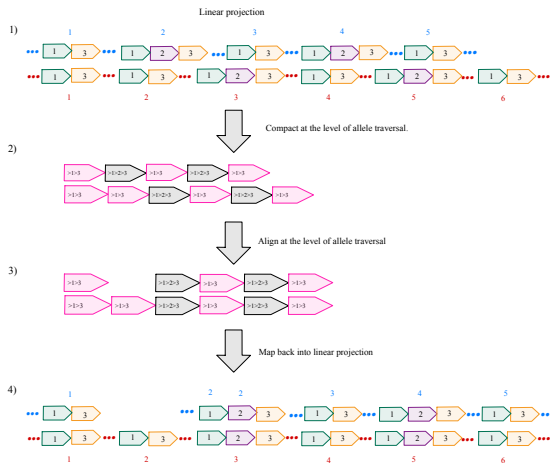
Alignment at the Node Level



Alignment at any level lower than that of allele traversal

Context is important for untangling.

Alignment at the Level of Allele Traversals



Summary/Too Long Didn't Listen (TLDL)

- unroll and align

Acknowledgements

Nadia Pisanti

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